PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Electrical load list, cables and hazardous area

PVC-V1-EE-LL-001

Electrical load list of the R-101 bay generated from the OTS model: 13 loads, maximum demand 841 kVA during degassing; agitator M-101 450 kW at 6.6 kV for a 338 kW peak; hazardous area for vinyl chloride (IIA, T2, cited sources), motors IIB T3. Checks: 32 ✓ / 2 ✗; 6 engineering queries.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
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The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Purpose and basis

This document gives the electrical load list of the R-101 bay with the maximum demand, the cable-sizing method, the hazardous-area classification for vinyl chloride with the Ex selection of the motors, and what a power failure does to the batch. Absorbed powers come from the R-101 simulator model and its reference batch; loads without a model power are labelled assumed; the VCM flammability data are cited.

Model powers	M-101 agitator: step() P.Qag = K.agitKW (290 kW) × (0.9 + 0.25 X) × level factor; P-101A/B 90 kW and P-103 55 kW from econ()
Reference batch	Reference batch pvBatch({}): start of the reaction (57 °C, 60 t water + 51 t VCM, initiator in), dt 0.25 s, noise off, SIS on, stops at the PD-100 end point.
Rating rule	rated = next IEC size ≥ absorbed / η_drive × (1 + margin); margins 15 % agitator, 10 % pumps (typical); η_motor IE3 4-pole (IEC 60034-30-1, typical)
Voltage rule (typical)	≤ 200 kW at 400 V, above at 6.6 kV; LV ≤ 55 kW DOL, larger LV motors soft starter
Maximum demand	841 kVA (738 kW) during degassing

2 Electrical load list

Tag	Service	Src	Absorbed kW	Rated kW	V	Duty	Starter	η · cos φ	Input kW	kVA	I, A	Bus	Area / Ex
M-101	R-101 agitator	model	338.1	450	6600	Running	DOL (MV)	0.960 · 0.88	363.1	412.6	36.1	Essential (R02)	Zone 2 (seal, secondary grade) · Ex db IIB T3, EPL Gb
P-101A	Jacket circulation pump (duty)	model	90.0	110	400	Running	Soft starter	0.954 · 0.87	94.3	108.4	156.5	Essential (R02)	Zone 2 · Ex ec IIB T3
P-101B	Jacket circulation pump (standby)	model	90.0	110	400	Standby	Soft starter	0.954 · 0.87	94.3	108.4	156.5	Essential (R02)	Zone 2 · Ex ec IIB T3
P-103	VCM charge pump	model	55.0	75	400	Intermittent	Soft starter	0.950 · 0.87	57.9	66.5	96.1	Normal	Zone 2 (VCM pump: zone 1 around the seal) · Ex db IIB T3
VP-101	Vacuum pump (liquid ring)	assumed	75.0	90	400	Intermittent	Soft starter	0.952 · 0.87	78.8	90.6	130.7	Normal	Zone 2 · Ex ec IIB T3
C-101	VCM recovery compressor (shared by R-101…R-104)	assumed	210.1	250	6600	Intermittent	DOL (MV)	0.960 · 0.88	218.8	248.7	21.8	Normal	Zone 2 · Ex db IIB T3
P-102	Slurry pump D-102 → T-201	assumed	18.2	22	400	Intermittent	DOL	0.930 · 0.85	19.6	23.0	33.2	Normal	Zone 2 · Ex ec IIB T3
TK-107 M	PVA pot mixer	assumed	1.8	2.2	400	Intermittent	DOL	0.867 · 0.82	2.1	2.5	3.7	Normal	Safe area (no VCM)
D-102 M	Blowdown / slurry tank agitator	assumed	37.0	45	400	Running	DOL	0.942 · 0.85	39.3	46.2	66.7	Normal	Zone 2 · Ex db IIB T3
P-104	Initiator dosing pump (positive displacement)	assumed	1.1	1.5	400	Intermittent	DOL	0.853 · 0.82	1.3	1.6	2.3	Normal	Zone 2 · Ex db IIB T3
D-104 CH	Initiator pot chiller package	assumed	5.5	7.5	400	Running	DOL	0.904 · 0.82	6.1	7.4	10.7	Essential	Safe area or Ex package
UPS-101	UPS: SIS, DCS, instruments, solenoids, gas detection	assumed	6.3	—	230	Running	—	— · 0.9	6.3	7.0	30.5	UPS (charger on essential)	Safe area (control room / substation)
LTG-101	Lighting and small power, R-101 bay	assumed	10.0	—	400	Running	—	— · 0.95	10.0	10.5	15.2	Normal (escape lighting on UPS)	Zone 2 luminaires Ex ec / Ex db

2.1 Notes per load

• **M-101**: step(): P.Qag = K.agitKW (290 kW) × (0.9 + 0.25 X) × level factor; 290 kW at the design state (X 0.26), peak 338 kW at X 0.848 (reference batch); gearbox η 0.97 (assumed)

• **P-101A**: econ(): 90 kW while P-101A or B runs; 130 kg/s (K.Fc)

- **P-101B**: Auto-start on P-101A trip recommended (HAZOP R01); same rating as P-101A
- **P-103**: econ(): 55 kW while P-103 runs (VCM charge)
- **VP-101**: Assumed 75 kW for 3240 m³/h (K.Svac 0.9 m³/s) to 0.03 bar a; typical for a liquid-ring pump of that size
- **C-101**: Computed: residual VCM 7.76 t at the end point (reference batch) degassed in 1 h (assumed) = 2.16 kg/s; 1.25 → 6.05 bar a (VCM condensing at 40 °C), isentropic 73.0 kJ/kg, η 0.75 (assumed)
- **P-102**: Computed: 121.3 m³ batch in 1 h = 121 m³/h, head 30 m, ρ 1100 kg/m³, η 0.6 (assumed)
- **TK-107 M**: Assumed 1.8 kW absorbed (2.2 kW motor in the Mechanical equipment list)
- **D-102 M**: Assumed 37 kW absorbed (45 kW motor in the Mechanical equipment list); keeps the slurry suspended
- **P-104**: Assumed 1.1 kW (60-s dose from D-104)
- **D-104 CH**: Assumed 5.5 kW; peroxide storage cooling (HAZOP R05)
- **UPS-101**: Sum of the UPS loads (section 4), assumed
- **LTG-101**: Assumed 10 kW

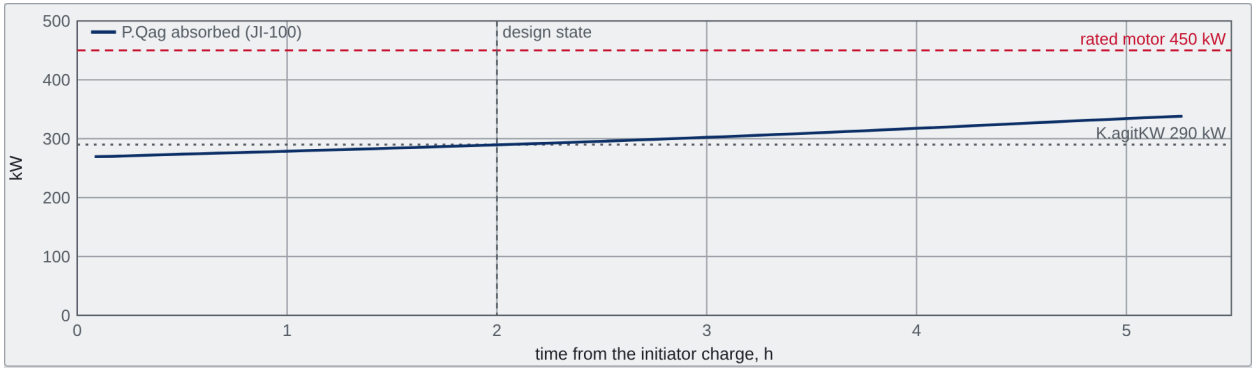


Figure 1 · Agitator M-101 absorbed power over the reference batch, the model base power K.agitKW and the rated motor.

Worked example, M-101. The model's base power K.agitKW = 290 kW is scaled in step() by (0.9 + 0.25 *X*) and the level: 290 kW at the design state (2.0 h, *X* 0.26), rising to 338 kW at the end point (*X* 0.848, level 83.9 %). Required motor output = 338.1 / 0.97 (gearbox, assumed) × (1 + 15 %) = 400.9 kW → next IEC size **450 kW**; above 200 kW the motor goes to 6.6 kV (typical practice), so *I* = 412.6 kVA / (√3 × 6.6 kV) = 36.1 A. Electrical input at the peak 363.1 kW (η_{motor} 0.960, IE3 typical).

2.2 Maximum demand by batch phase

Batch phase	Loads running	Input, kW	kVA
Evacuation	VP-101, D-102 M, D-104 CH, UPS-101, LTG-101	140	162
Water charge	M-101, P-101A, TK-107 M, D-102 M, D-104 CH, UPS-101, LTG-101	521	595
VCM charge	M-101, P-101A, P-103, D-102 M, D-104 CH, UPS-101, LTG-101	577	659

Batch phase	Loads running	Input, kW	kVA
Heat-up	M-101, P-101A, D-102 M, D-104 CH, UPS-101, LTG-101	519	592
Reaction	M-101, P-101A, D-102 M, P-104, D-104 CH, UPS-101, LTG-101	520	594
Degassing (maximum)	M-101, P-101A, C-101, D-102 M, D-104 CH, UPS-101, LTG-101	738	841
Discharge	M-101, P-102, D-102 M, D-104 CH, UPS-101, LTG-101	444	507

Connected load 992 kW; conventional estimate with demand factors (running 1, intermittent 0.3, standby 0): 633 kW / 722 kVA. Essential running load: M-101, P-101A, D-104 CH, UPS-101 = 535 kVA.

3 Cable sizing

$I_b = P / (\sqrt{3} \cdot V \cdot \cos\varphi \cdot \eta)$; a size passes when $Iz \cdot Ct \cdot Cg \geq I_b$, the running voltage drop $\Delta V\ \% = \sqrt{3} \cdot I \cdot L \cdot (R \cos\varphi + X \sin\varphi)/V \times 100 \leq 5\ \%$ and the starting drop ≤ 15 % (typical limits). Cu XLPE multicore on tray, IEC 60364-5-52 method E ratings and impedances (typical); route lengths and derating factors assumed (Ct 0.91 for 40 °C, Cg 0.80 for a grouped tray).

Load	P, kW	V	Starter	L, m	I_b, A	Start, A	Size, mm²	ΔV run, %	ΔV start, %
M-101	450	6600	DOL (MV)	150	46.6	303	35	0.12	0.3
P-101A	110	400	Soft starter	150	191.3	670	95	3.17	7.1
P-103	75	400	Soft starter	150	131.0	458	50	4.00	7.5
VP-101	90	400	Soft starter	150	156.8	549	70	3.44	7.0
C-101	250	6600	DOL (MV)	250	25.9	168	35	0.11	0.3
P-102	22	400	DOL	150	40.2	261	16	3.38	8.9
TK-107 M	2.2	400	DOL	150	4.5	29	2.5	2.26	5.5
D-102 M	45	400	DOL	150	81.1	527	25	4.39	12.3
P-104	1.5	400	DOL	150	3.1	20	2.5	1.57	3.8
D-104 CH	7.5	400	DOL	150	14.6	95	4	4.63	11.5

4 Hazardous-area classification

Property	Value	Source	Meaning for R-101
Flash point	−78 °C (closed cup)	ICSC 0082; INRS FT 184; CAMEO −110 °F	A gas at ambient: any release is flammable at once
Flammable limits	3.6 – 33 vol % in air (IISc sheet: 3.8–31); LEL ≈ 94 g/m³ at 20 °C (computed)	ICSC 0082; CAMEO; INRS FT 184; DOE-HDBK-1092	Wide range; gas detectors alarm at 20 % LEL (typical)
Auto-ignition temperature	415 °C and 472 °C	IISc ENVIS sheet (ignition temperature); ICSC 0082, CAMEO, INRS FT 184, DOE-HDBK-1092 (472 °C)	Take the lower value: 415 °C → temperature class T2 (≤ 300 °C)
Gas group	NEC Class I, Group D → IEC IIA (Group D corresponds to IIA, NEC 505 / IEC 60079-20-1 practice)	DOE-HDBK-1092 Table 5-5	Equipment group IIA, IIB or IIC is suitable
Boiling point	−13 °C (model DIPPR curve: −14.0 °C)	ICSC 0082; model PsV	A release from 57 °C flashes: adiabatic flash fraction ≈ 25 % (computed below)
Vapour density	2.2 (air = 1)	ICSC 0082; CAMEO 2.21	Heavier than air: collects at grade, in trenches and pits

Why the extents reach grade. Liquid VCM in R-101 is at 57.0 °C and 8.62 bar g (design state). Released to the atmosphere it boils at −14.0 °C (model DIPPR curve): the adiabatic flash fraction $1 - \exp(-c_p \Delta T/\lambda) = 1 - \exp(-1.35 \times 71.0 / 334) \approx$ **25 %** flashes at once and the rest forms a cold boiling pool and aerosol; the vapour, 2.2 × heavier than air, spreads along the ground and into trenches. Its lower limit, 3.6 vol %, is about 94 g/m³ — a few kilograms make a large flammable cloud.

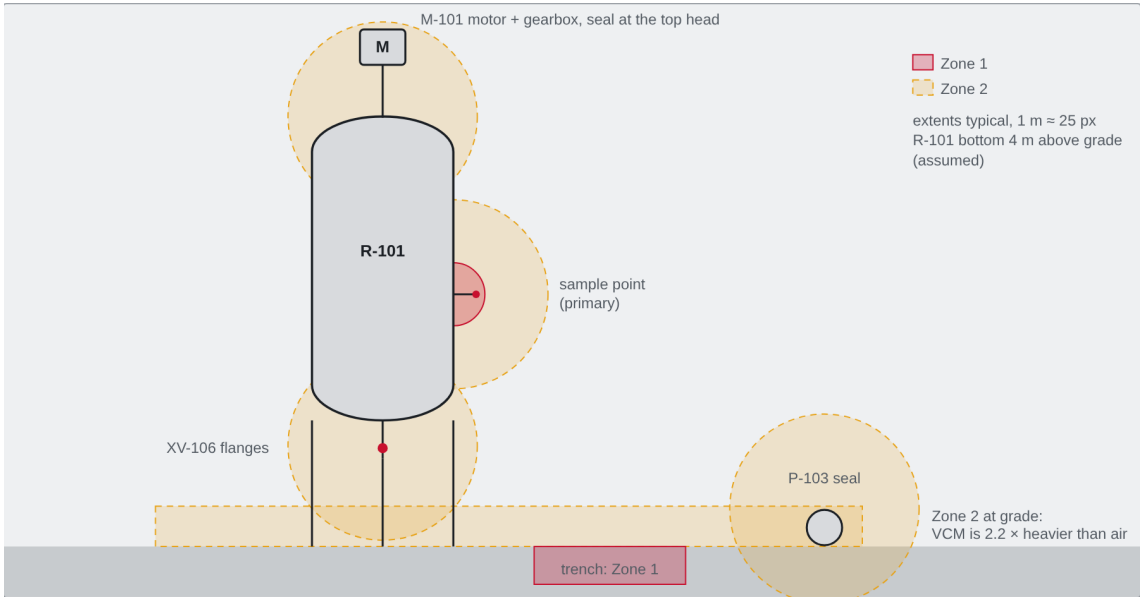


Figure 2 · Elevation of the R-101 bay with typical zone extents (illustration, not a calculated classification).

Source of release	Grade	Ventilation	Zone	Extent (typical)
Agitator M-101 shaft seal (double mechanical seal with barrier fluid)	Secondary	Natural, open structure (assumed): medium degree, good availability	Zone 2	3 m around the seal housing and down to grade
Flanges, valve stems and instrument connections on R-101 and the charge lines	Secondary	As above	Zone 2	1–3 m around each source
Sample points on R-101 / D-102 (opened during operation)	Primary	As above	Zone 1 + Zone 2	Zone 1 within 1 m, Zone 2 to 3 m
Manway opened for cleaning (after degassing and purge)	Secondary (only if the purge fails)	As above	Zone 2	3 m around the manway
PSV-100 / XV-108 discharge	None at R-101 (closed to D-103)	—	—	Zone at the final vent of D-103 / recovery, not at R-101
Degassing and inert vent	None at R-101 (closed to D-106 recovery)	—	—	Flanges only (row 2)
Trenches, pits and sumps below grade in the bay	Release collects (VCM vapour density 2.2 × air)	Poor (no air movement)	Zone 1	The whole trench or pit
P-103 VCM charge pump seal	Secondary	As above	Zone 2 (Zone 1 close to the seal)	3 m, extended at grade

Motor	Location	Zone	EPL	Protection	Group · T	Notes
M-101	R-101 agitator	Zone 2 (seal as a source)	Gb	Ex db (flameproof) or Ex eb (increased safety)	IIB T3	MV Ex motor: check stator-winding discharge and start-up purge (Ex eb / Ex p) requirements.
P-101A	Jacket circulation pump (duty)	Zone 2	Gc	Ex ec (non-sparking; formerly Ex nA)	IIB T3	Soft starter / VSD duty must be in the motor certificate (temperature at reduced speed).

Motor	Location	Zone	EPL	Protection	Group · T	Notes
P-101B	Jacket circulation pump (standby)	Zone 2	Gc	Ex ec (non-sparking; formerly Ex nA)	IIB T3	Soft starter / VSD duty must be in the motor certificate (temperature at reduced speed).
P-103	VCM charge pump	Zone 2 (seal as a source)	Gb	Ex db (flameproof) or Ex eb (increased safety)	IIB T3	Soft starter / VSD duty must be in the motor certificate (temperature at reduced speed).
VP-101	Vacuum pump (liquid ring)	Zone 2	Gc	Ex ec (non-sparking; formerly Ex nA)	IIB T3	Soft starter / VSD duty must be in the motor certificate (temperature at reduced speed).
C-101	VCM recovery compressor (shared by R-101…R-104)	Zone 2 (seal as a source)	Gb	Ex db (flameproof) or Ex eb (increased safety)	IIB T3	MV Ex motor: check stator-winding discharge and start-up purge (Ex eb / Ex p) requirements.
P-102	Slurry pump D-102 → T-201	Zone 2	Gc	Ex ec (non-sparking; formerly Ex nA)	IIB T3	
TK-107 M	PVA pot mixer	safe area or package	—	—	—	
D-102 M	Blowdown / slurry tank agitator	Zone 2 (seal as a source)	Gb	Ex db (flameproof) or Ex eb (increased safety)	IIB T3	
P-104	Initiator dosing pump (positive displacement)	Zone 2 (seal as a source)	Gb	Ex db (flameproof) or Ex eb (increased safety)	IIB T3	
D-104 CH	Initiator pot chiller package	safe area or package	—	—	—	

5 Power failure and UPS

A power failure is a process event before it is an electrical one. What the R-101 batch logic does (read from step(), tripConditions() and bmsTrip() in the model):

• **Motors stop.** The model stops every motor in MOTORS (M101, P101A, P101B, P103). Cooling water and instrument air are site utilities and are kept in the model (whether their own pumps and compressors share the supply is a site question, HAZOP R09).

• **Agitation and jacket circulation decay.** The mixing state φ falls to zero with a 15-s time constant (jacket heat transfer drops to 30 % of its agitated value, the baffles to 40 %), and the P-101 flow decays with a 4-s time constant.

• **The SIS trips.** While the batch reacts, XS-101 (agitator stopped, 15 s) and FSLL-110 (jacket circulation below 120 m³/h, 30 s) are armed; either one gives Trip A: short-stop through XV-104, charges XV-101/102 closed, forced cooling (TV-110B, FV-120, FV-130 to 100 %, TV-110A closed). The trip is latched and can be reset only when the high-high conditions have cleared.

• **The short-stop needs no motor.** Its motive force is the nitrogen pressure in the short-stop pot D-105, and XV-104 is a pneumatic valve; it needs only the SIS output (UPS-backed) and instrument air (dedicated receiver, HAZOP R39). The cooling valves are pneumatic and fail open on loss of air.

• **But it mixes slowly without the agitator.** In step() the short-stop reaches the reaction with a 20-s time constant when the batch is agitated and 400 s when it is not — the reason HAZOP R02 asks for M-101 and P-101A/B on essential power, and the load list puts them on the essential bus.

• **Restart.** Restarting the agitator on a batch that may still react is a written, deliberate step (HAZOP R38), not an automatic re-acceleration; the essential bus must be able to start M-101 (DOL at 6.6 kV) after a blackout.

UPS. The SIS logic solver, the DCS, the field instruments, the on/off valve solenoids, gas detection and escape lighting are on a UPS, so the protection and the operator's view survive the failure. With every protection failed, the Relief & vent tab's power-failure run reaches the PSV-100 set 77:32 after the failure — the horizon against which the UPS autonomy is checked in part 5.

UPS load (assumed)	kW
SIS logic solver and I/O (separate, P&ID-001 note 8)	1.50
DCS controllers and I/O for R-101	2.00
Operator stations (2)	1.00
Field instruments, 4–20 mA loops and analysers	0.80
Solenoids of the on/off valves (XV-101 … XV-109, 8 × 15 W)	0.12
Fixed gas detectors (HAZOP R21)	0.30
Escape lighting and alarm horn	0.60
Total → 8.78 kVA with cos φ 0.9 and growth 1.25 → 10 kVA UPS, 30 min (typical)	6.32

6 Checks against the model and the other documents

32 ✓ · 2 ✗ (computed in the OTS page when this document was generated).

Check	Value	Criterion	✓/✗	Source
Rated ≥ absorbed peak: M-101	450 kW vs 338.1 kW	rated ≥ absorbed / η_{drive}	✓	Model
Rated ≥ absorbed peak: P-101A	110 kW vs 90.0 kW	rated ≥ absorbed / η_{drive}	✓	Model

Check	Value	Criterion	✓/✗	Source
Rated ≥ absorbed peak: P-101B	110 kW vs 90.0 kW	rated ≥ absorbed / η_{drive}	✓	Model
Rated ≥ absorbed peak: P-103	75 kW vs 55.0 kW	rated ≥ absorbed / η_{drive}	✓	Model
Rated ≥ absorbed peak: VP-101	90 kW vs 75.0 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: C-101	250 kW vs 210.1 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: P-102	22 kW vs 18.2 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: TK-107 M	2.2 kW vs 1.8 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: D-102 M	45 kW vs 37.0 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: P-104	1.5 kW vs 1.1 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Rated ≥ absorbed peak: D-104 CH	7.5 kW vs 5.5 kW	rated ≥ absorbed / η_{drive}	✓	Assumed load
Load list = model: M-101 at the design state	290.1 kW vs P.Qag 290.1 kW	equal	✓	step() / snapDesign
Load list = model: P-101 (econ())	90 kW vs 90 kW	equal	✓	econ() on the design state
Load list = model: P-103 (econ())	55 kW vs 55 kW	equal	✓	econ() with P-103 on
Model "full load" K.agitKW covers the agitator peak	290 kW vs 338 kW	K.agitKW ≥ peak	✗	model.js K, reference batch
Every model motor (MOTORS) in the load list	M-101, P-101A, P-101B, P-103	all listed	✓	model.js MOTORS
Every motor on PFD-100 in the load list	P-103, P-104, M-101, VP-101, C-101, P-101A, P-101B, P-102 (shared T-201, S-201, D-201 outside the R-101 area)	all listed	✓	PFD-100 equipment
Motor temperature class suits VCM	T3 = 200 °C vs lowest AIT 415 °C	max. surface temperature < AIT (and ≤ T2)	✓	EE_VCM (cited)
Equipment gas group covers VCM	IIB equipment for a IIA gas	IIB ⊇ IIA	✓	EE_VCM (cited)
Model boiling point of VCM vs the safety card	−14.0 °C vs −13 °C	within 2 K	✓	PsV (DIPPR) in model.js, ICSC 0082
UPS rating covers the SIS / DCS loads	10 kVA vs 8.78 kVA (incl. growth)	rating ≥ load	✓	Section 4 (assumed loads)
SIS, DCS and solenoids are on the UPS	UPS (charger on essential)	on UPS	✓	Load list
UPS autonomy covers a power failure with the SIS failed (time to PSV-100 lift)	30 min vs 77.5 min	autonomy ≥ time	✗	Relief & vent run 3.4
M-101 and P-101A/B on essential power (HAZOP R02)	M-101: Essential, P-101A: Essential, P-101B: Essential	essential	✓	Load list
Cable sizing defaults: M-101	35 mm², ΔV run 0.1 %, start 0.3 %	I ≤ Iz·Ct·Cg, ΔV ≤ 5 % / 15 %	✓	Calculator 3 (assumed route)
Cable sizing defaults: P-101A	95 mm², ΔV run 3.2 %, start 7.1 %	I ≤ Iz·Ct·Cg, ΔV ≤ 5 % / 15 %	✓	Calculator 3 (assumed route)
EM_EQL motor power covers the load-list requirement: M-101	450 kW (EM_EQL) vs required 400.9 kW, rated 450 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: P-101A	110 kW (EM_EQL) vs required 99.0 kW, rated 110 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: P-101B	110 kW (EM_EQL) vs required 99.0 kW, rated 110 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: P-103	75 kW (EM_EQL) vs required 60.5 kW, rated 75 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: VP-101	90 kW (EM_EQL) vs required 82.5 kW, rated 90 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: C-101	250 kW (EM_EQL) vs required 231.1 kW, rated 250 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: P-102	22 kW (EM_EQL) vs required 20.0 kW, rated 22 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL
EM_EQL motor power covers the load-list requirement: P-104	1.5 kW (EM_EQL) vs required 1.2 kW, rated 1.5 kW here	EM_EQL ≥ absorbed / η × (1 + margin)	✓	Mechanical tab EM_EQL

7 Engineering queries

#	Subject	Finding	Raised against	Proposed
EE-01	Agitator power: model "full load" vs the batch peak	K.agitKW is 290 kW, but step() scales it by (0.9 + 0.25 X) and the level, so the reference batch draws 290 kW at the design state and 338 kW at the end point (X 0.848). With gearbox η 0.97 and 15 % margin (assumed) M-101 needs 401 kW → 450 kW IEC.	Model (K.agitKW label), Mechanical (M-101 motor rating)	State M-101 as 450 kW rated / 338 kW absorbed peak in the equipment list and datasheet; keep K.agitKW as the base power in the model notes, not "full load".
EE-02	Essential power supply not shown	HAZOP R02 asks for M-101 and P-101A/B on essential power; the load list puts M-101, P-101A, D-104 CH, UPS-101 (running loads) on an essential bus of 535 kVA running load (M-101 DOL start on top). No standby generator or essential switchboard appears on PFD-100 or P&ID-001.	Electrical single-line (not yet issued), P&ID-001, HAZOP R02	Add an essential board fed by a standby generator sized for the essential load (≥ 669 kVA with 25 % margin, typical) and for the M-101 start; prove by test that the short-stop is mixed after a power failure.
EE-03	UPS autonomy vs the power-failure horizon	With the SIS failed, a power failure at the design state reaches the PSV-100 set after 77:32 (Relief & vent run 3.4); a typical 30-min UPS keeps the SIS and DCS alive for less than that if the essential supply also fails.	Electrical (UPS specification), SRS of SIF	Specify UPS autonomy ≥ 80 min for the SIS, DCS and the short-stop solenoid, or put the UPS charger on the essential (generator) bus and state it in the SRS.
EE-04	VCM auto-ignition temperature differs between sources	ICSC 0082, NOAA CAMEO, INRS FT 184 and DOE-HDBK-1092 give an auto-ignition temperature of 472 °C (T1); the IISc ENVIS data sheet gives an ignition temperature of 415 °C (T2). The temperature class must be chosen on the lower value.	PVC-V1-EE-LL-001 (hazardous-area classification)	Classify the R-101 area IIA T2; specify equipment T3 (200 °C) or better so the choice does not depend on the source.
EE-05	C-101 recovery compressor duty	C-101 is shared by R-101…R-104 and has no power in the model. The load list estimates 210 kW from the residual VCM at the end point (7.76 t) degassed in 1 h (assumed) from 1.25 to 6.05 bar a.	Process (degassing rate, compressor sizing), Mechanical	Confirm the degassing schedule of the four reactors (staggered or not) and the C-101 design flow; resize the motor and its feeder.
EE-06	Hazardous-area extents are typical only	Zone extents use typical distances: the ventilation of the reactor bay, the layout and the release rates of the sources (IEC 60079-10-1) are not yet defined.	Layout (step 1.6), Safety	Calculate release rates and the ventilation degree per IEC 60079-10-1 once the layout of the reactor bay is fixed; issue the hazardous-area drawing.

8 Sources and references

• ICSC 0082 Vinyl chloride (ILO/WHO International Chemical Safety Cards): flash point −78 °C (c.c.), auto-ignition temperature 472 °C, explosive limits 3.6–33 vol % in air, boiling point −13 °C, relative vapour density 2.2. https://training.itcilo.org/actrav_cdrom2/en/osh/ic/75014.htm (accessed 27 October 2026)

• NOAA CAMEO Chemicals, Vinyl chloride datasheet: flash point −110 °F (−78.9 °C), LEL 3.6 %, UEL 33 %, auto-ignition temperature 882 °F (472 °C), vapour density 2.21 (air = 1). https://cameochemicals.noaa.gov/report?key=CH1692 (accessed 27 October 2026)

• INRS, Fiche toxicologique n° 184 Chlorure de vinyle (June 2024): point d'éclair −78 °C (closed cup), auto-ignition 472 °C, LIE 3.6 %, LSE 33 %, boiling point −14 to −13 °C. https://inrs.fr/dam/ficheTox/FicheFicheTox/FICHETOX_184-1.pdf (accessed 27 October 2026)

• US DOE, DOE-HDBK-1092-98 Electrical Safety, Table 5-5 (Class I hazardous atmospheres): vinyl chloride: Group D, AIT 882 °F / 472 °C, LFL 3.6 %, UFL 33.0 %. https://nuclearpowerradiation.tpub.com/hdbk1092/Table-5-5-Class-I-Division-1-And-Class-I-Division-2-Summary-Of-Selected-Hazardous-Atmospheres-Continued-97.htm (accessed 27 October 2026)

• IISc Energy & Wetlands Research Group, ENVIS hazardous-substance data sheet (vol. 369): vinyl chloride: ignition temperature 415 °C, auto-ignition 472 °C, flash point −77 to −78 °C, explosion limits 3.8–31 vol %. https://wgbis.ces.iisc.ac.in/energy/HC270799/HDL/ENV/enven/vol369.htm (accessed 27 October 2026)

• Standards quoted as typical practice: IEC 60079-10-1 (classification), IEC 60079-0/-1/-7 (Ex equipment, EPL, Ex db, Ex eb, Ex ec), IEC 60034-30-1 (motor efficiency classes), IEC 60364-5-52 (cable current ratings), NEC Article 500 (class/group).

• Model: R-101 OTS model (model.js: K, step(), econ(), MOTORS, tripConditions(), bmsTrip()); reference batch and design state (tabs/eelec/tools/compute.js); Relief & vent run 3.4 (power failure, SIS off).

• Other OTS documents: PFD-100, PVC-PID-001/002, HAZOP report PVC-V1-SA-HZP-001 (R01, R02, R05, R09, R21, R38, R39), SRS PVC-V1-IN-SRS-001, relief load basis PVC-V1-PE-RLB-001.